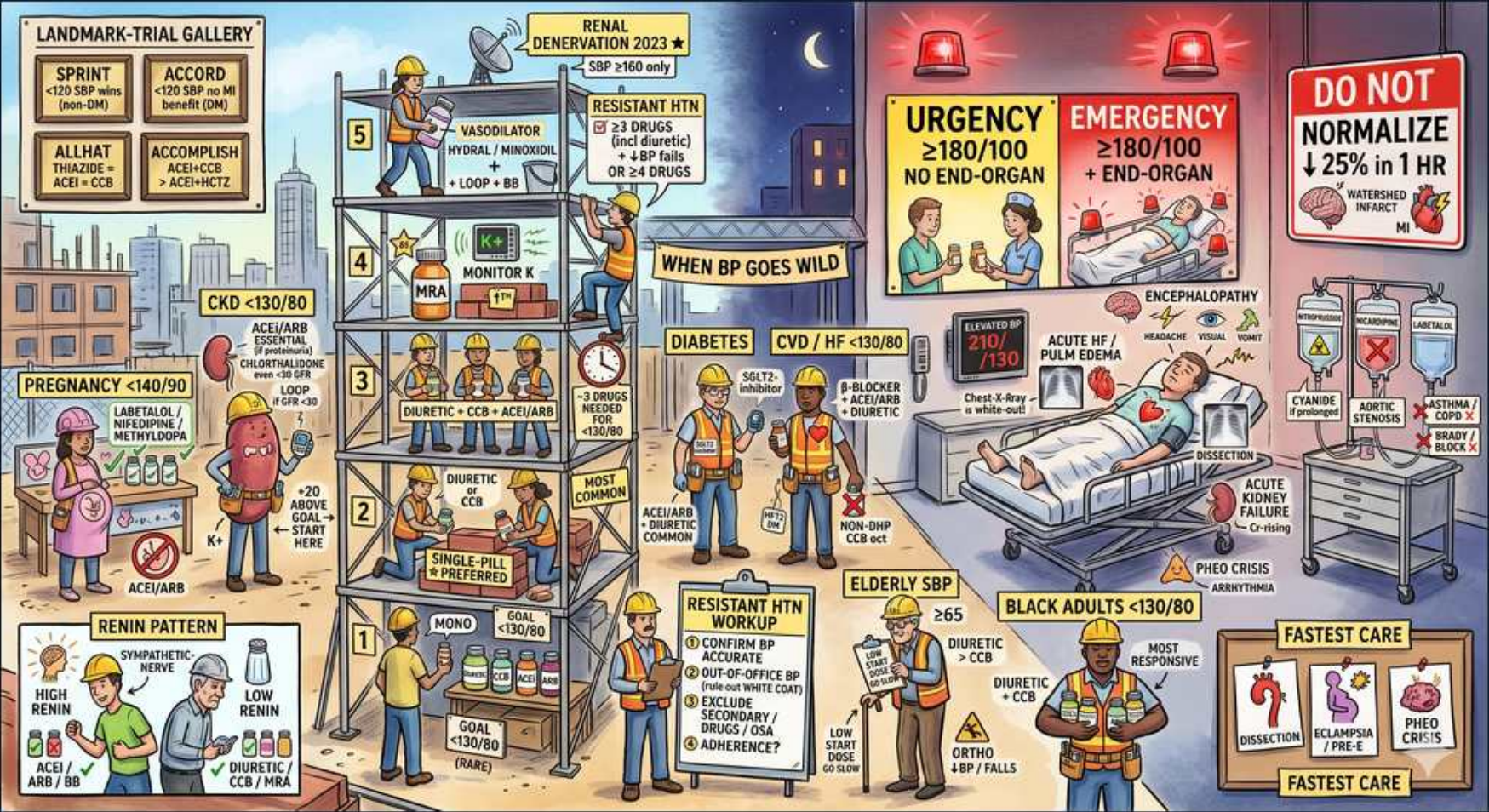


Hypertension — Urgency, Emergency & Special Populations



1. Urgency $\geq 180/100$

WHAT YOU SEE

The URGENCY $\geq 180/100$ panel marked 'NO end-organ' on a calm beige background — the absence of any injured-organ icon is the cue it is managed with oral drugs and outpatient follow-up.

WHAT IT ENCODES

Hypertensive URGENCY: BP $\geq 180/120$ with NO acute end-organ damage. Manage with ORAL agents (restart/up-titrate the patient's regimen — e.g. amlodipine, an ACEi/ARB, or a long-acting agent) and arrange outpatient follow-up in 1–7 days; the goal is gradual reduction over days, not minutes. No ICU, no IV drips, no continuous monitoring required.

BOARD TRAP

WRONG answer = admit to ICU and start an IV nitroprusside drip. In urgency there is no end-organ damage, so rapid IV lowering only risks hypoperfusion (stroke, MI, AKI). Oral therapy with close follow-up is correct; the discriminator is the ABSENCE of acute target-organ injury.

2. Emergency ≥ 180 + organ

WHAT YOU SEE

The EMERGENCY $\geq 180/100$ panel on an alarm-RED background with a flashing siren, '+END-ORGAN' — the red siren plus damaged-organ tag marks it as the ICU/IV-drip emergency.

WHAT IT ENCODES

Hypertensive EMERGENCY: severe BP (usually $>180/120$) WITH acute, ongoing end-organ damage. Requires ICU admission, an arterial line, and titratable IV agents — IV nicardipine or clevidipine (CCBs), labetalol (combined α/β), or nitroprusside. General target: lower MAP by ~ 10 – 20% in the first hour (max $\sim 25\%$), then toward $160/100$ – 110 over the next 2–6 hours.

BOARD TRAP

WRONG answer = define emergency by the BP number alone (e.g. 'BP $220/120$ with no symptoms = emergency'). It is the presence of ACUTE end-organ damage — encephalopathy, ACS, pulmonary edema, AKI, dissection, eclampsia — not the absolute number, that defines an emergency. The same $220/120$ with no organ injury is urgency.

3. Do not normalize fast

WHAT YOU SEE

DO NOT NORMALIZE, $\downarrow 25\%$ in 1HR, watershed infarct warning

WHAT IT ENCODES

Do NOT normalize BP acutely. Lower MAP by ~ 10 – 20% (never more than $\sim 25\%$) in the first hour. Chronic HTN shifts cerebral/renal autoregulation curves rightward, so a 'normal' BP can fall below the autoregulatory floor and cause watershed cerebral, renal, and coronary ischemia.

BOARD TRAP

WRONG answer = drive BP rapidly to $<140/90$ to 'protect the brain.' Overshoot below the autoregulatory threshold precipitates watershed (border-zone) cerebral infarction, AKI, and MI. Exceptions that DO warrant faster/lower targets are aortic dissection (SBP <120 , HR <60 within ~ 20 min) — the discriminator is the diagnosis, not the BP value.

4. End-organ targets

WHAT YOU SEE

Encephalopathy, acute HF/pulm edema, AKI, ICU bed

WHAT IT ENCODES

Acute end-organ targets in hypertensive emergency: hypertensive encephalopathy (headache, confusion, seizures, papilledema), acute pulmonary edema/HF, acute kidney injury, grade III/IV retinopathy, ACS/MI, aortic dissection, and eclampsia. Tailor the IV agent to the organ — e.g. nitroglycerin/clevidipine for pulmonary edema, labetalol/nicardipine for encephalopathy.

BOARD TRAP

WRONG answer = call asymptomatic severe HTN an 'emergency' because of an elevated creatinine that is actually chronic. Distinguish ACUTE injury (new rise in Cr, new neuro deficit, fresh retinal hemorrhages) from chronic baseline changes; only acute, evolving damage upgrades urgency to emergency.

5. Aortic dissection

WHAT YOU SEE

AORTIC dissection icon, esmolol then nitroprusside

WHAT IT ENCODES

Aortic dissection is the exception requiring FAST, AGGRESSIVE lowering: target SBP <120 mmHg and HR <60 within ~ 20 minutes. Give an IV beta-blocker FIRST (esmolol or labetalol) to blunt dP/dt (the shear/pulsatile force on the aortic wall), THEN add a vasodilator (nitroprusside) if SBP remains high.

BOARD TRAP

WRONG answer = start nitroprusside (a pure vasodilator) first. Vasodilation alone triggers reflex tachycardia and increased dP/dt, propagating the dissection. Beta-blockade MUST precede the vasodilator. The discriminator: control heart rate/contractility before pressure.

6. Pheo crisis

WHAT YOU SEE

PHEO CRISIS, phentolamine

WHAT IT ENCODES

Pheochromocytoma crisis: catecholamine surge → give IV alpha-blockade FIRST (phentolamine, or nicardipine); beta-blockade may be added only AFTER adequate alpha-blockade to control reflex tachycardia. Suspect with paroxysmal HTN plus headache, palpitations, and diaphoresis.

BOARD TRAP

WRONG answer = give a beta-blocker (e.g. propranolol/labetalol) up front. Unopposed alpha stimulation from beta-blockade causes severe vasoconstriction and a hypertensive crisis. Alpha BEFORE beta — same rule as elective pheo surgical prep (phenoxylbenzamine first).

7. Eclampsia / pregnancy

WHAT YOU SEE

ECLAMPSIA, labetalol/hydralazine, MgSO₄

WHAT IT ENCODES

Severe preeclampsia/eclampsia: treat BP with IV labetalol, IV hydralazine, or oral immediate-release nifedipine (target $<160/110$), and give IV magnesium sulfate for seizure prophylaxis/treatment. Definitive treatment is delivery of the fetus/placenta.

BOARD TRAP

WRONG answer = use an ACEi/ARB or nitroprusside. ACEi/ARBs are teratogenic (renal dysgenesis, oligohydramnios) and contraindicated; nitroprusside risks fetal cyanide toxicity. Also WRONG = treat seizures with standard antiepileptics — magnesium is the agent of choice in eclampsia.

8. Resistant HTN definition

WHAT YOU SEE

The RESISTANT HTN placard showing '≥3 drugs incl. a diuretic OR ≥4 drugs' — the bottle count IS the definition.

WHAT IT ENCODES

Resistant HTN: BP above goal despite adherence to ≥3 antihypertensives at optimal/max doses from different classes, ONE of which is a diuretic — OR BP controlled but requiring ≥4 agents. Prompts evaluation for secondary causes and pseudoresistance.

BOARD TRAP

WRONG answer = label as resistant when the patient is on three drugs but no diuretic, or on submaximal doses. A diuretic at adequate dose is mandatory to meet the definition; otherwise it's undertreated, not resistant.

9. Add spironolactone

WHAT YOU SEE

The resistant-workup desk with an MRA / spironolactone bottle as the 4th-line add-on (PATHWAY-2).

WHAT IT ENCODES

For true resistant HTN, the preferred 4th-line add-on is a mineralocorticoid receptor antagonist — spironolactone 25–50 mg/day (eplerenone if antiandrogen effects like gynecomastia occur). The PATHWAY-2 trial showed spironolactone superior to bisoprolol/doxazosin as the 4th agent.

BOARD TRAP

WRONG answer = escalate to a 4th agent before confirming adherence, proper BP technique, white-coat effect, and excluding secondary causes (especially primary aldosteronism). Also monitor K⁺ and creatinine after starting an MRA — hyperkalemia is the key adverse effect, particularly with concurrent ACEi/ARB or CKD.

10. Resistant workup

WHAT YOU SEE

The detective-style RESISTANT HTN WORKUP papers: confirm BP technique, out-of-office readings, adherence, exclude secondary causes — the checklist pictures ruling out pseudoresistance before adding drugs.

WHAT IT ENCODES

Resistant-HTN workup before escalating: confirm correct BP measurement technique, obtain out-of-office (home/ambulatory) readings to exclude white-coat effect, verify medication adherence, review interfering substances (NSAIDs, decongestants, OCPs, alcohol, stimulants, licorice), and screen for secondary causes (primary aldosteronism, OSA, renovascular, CKD).

BOARD TRAP

WRONG answer = jump straight to renal denervation or a 5th drug. Most apparent resistance is pseudoresistance (nonadherence, white-coat, bad technique, or interfering drugs); these must be excluded first. Primary aldosteronism is the most common identifiable secondary cause in resistant HTN.

11. Single-pill combo

WHAT YOU SEE

The SINGLE-PILL PREFERRED sign with goal <130/80 — one combined tablet shown to picture better adherence.

WHAT IT ENCODES

Single-pill (fixed-dose) combinations are preferred to improve adherence and reach goal faster. General ACC/AHA target is <130/80 for most adults. Rational pairings: ACEi/ARB + thiazide, or ACEi/ARB + dihydropyridine CCB (the ACCOMPLISH-trial combo, benazepril+amlodipine, outperformed ACEi+HCTZ).

BOARD TRAP

WRONG answer = combine an ACEi WITH an ARB (or add a direct renin inhibitor to either). Dual RAAS blockade increases hyperkalemia, hypotension, and AKI without added benefit (ONTARGET/ALTITUDE). Combine across mechanisms, not within the same pathway.

12. Diabetes / CKD

WHAT YOU SEE

The DIABETES & CKD pillars with an ACEi/ARB bottle starred as preferred — the renoprotective first choice for these compelling indications.

WHAT IT ENCODES

In diabetes and CKD with albuminuria, an ACEi or ARB is the compelling first-line choice for renoprotection (reduces intraglomerular pressure and slows progression of proteinuric nephropathy). Target generally <130/80.

BOARD TRAP

WRONG answer = combine ACEi + ARB for 'extra' kidney protection in diabetic nephropathy. Dual blockade worsens hyperkalemia and AKI without slowing CKD. Use ONE RAAS blocker; monitor K⁺ and creatinine (a rise up to ~30% is acceptable and expected).

13. Black adults

WHAT YOU SEE

BLACK ADULTS <130/80, thiazide/CCB most responsive

WHAT IT ENCODES

In Black adults WITHOUT CKD or albuminuria, initial therapy should be a thiazide diuretic or a calcium channel blocker, which are more effective than ACEi/ARB as monotherapy in this group (lower-renin physiology). Target <130/80.

BOARD TRAP

WRONG answer = start an ACEi/ARB as monotherapy. It is less effective for BP lowering and carries a higher risk of angioedema in Black patients. Exception: if CKD with albuminuria is present, an ACEi/ARB IS indicated for renoprotection regardless of race.

14. Elderly SBP

WHAT YOU SEE

The ELDERLY SBP panel with an orthostatic/falls caution (a stumbling elder figure) — the fall icon warns against over-aggressive lowering.

WHAT IT ENCODES

In older adults, treat isolated systolic hypertension (SBP-driven); the SPRINT trial supported intensive control (SBP ~120 by automated office measurement) even in fit elderly. Start low, go slow, and check standing BP at each visit.

BOARD TRAP

WRONG answer = push aggressively to a low SBP in a frail elder while ignoring symptoms. Watch for orthostatic hypotension, dizziness, and falls; an excessively low DBP may compromise coronary perfusion. Individualize the target by frailty and tolerance.